ANSWER KEY

25900139 **A**

Reg.No : .....
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
NA SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCMAT100 - APPLICABLE MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 1.25 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple type Questions

Answer any **25** questions

Each question carries **2** marks

1. B: $\begin{bmatrix} 9 & 3 & 6 \\ 18 & -3 & 24 \end{bmatrix}$
2. A:3 Elements
3. A:Both PQ and QP are defined.
4. B:3×2
5. B: $\begin{bmatrix} 0 & 1 & 0 \\ -3 & 6 & -2 \\ 7 & -3 & -3 \end{bmatrix}$
6. B: $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$
7. D: $\begin{bmatrix} 14 & 14 \\ 13 & 12 \end{bmatrix}$
8. D:Skew-symmetric matrix
9. A:$(3x-2)(x+4)$

10. C:3
11. C:$x + 1$
12. C:0
13. A:$2x + 1$
14. D:$1, 7$
15. B:$\frac{^{12}C_3 \times ^9C_3 \times ^6C_3 \times ^3C_3}{4!}$
16. B:8^8
17. D:$\frac{(n-1)!}{n+1}$
18. B:24
19. B:$\frac{m(m-1)(m-2)}{6}$
20. D:2520
21. A:n + 1
22. A:45
23. C:$-x \sin x + \cos x$
24. For any answer, give full marks
25. A:0
26. C:$\sec^2 x$
27. B:$8x \sec^2(4x^2)$
28. B:$2 - 2x$
29. B:0
30. A:$\frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x))$

[2x25 = 50]